DATABASE MANAGEMENT SYSTEMS · PROJECT · PHASE 2

MUSE — Music Mood Intelligence & Recommendation Database System

Database Creation (DDL) and Sample Data (DML)

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DBMS: MySQL 8.0.46 Community Server (InnoDB, utf8mb4), MySQL Workbench 8.0 CE. Every figure and test result in this report was produced by running the delivered scripts on that server.

1. What Phase 2 delivers

Phase 1 designed the MUSE database (ER diagram and normalisation). Phase 2 creates it: the DDL script defines every table with its columns, data types, primary keys, foreign keys, unique keys, CHECK constraints, defaults and indexes, and the DML script fills it with enough sample data to test the design — 15,712 rows in 64 tables.

File (folder database/phase2/)	Size	Purpose
MUSE_Phase2_DDL.sql	59 KB	Creates database muse_db and all 64 tables with every key, constraint and index. Run first.
MUSE_Phase2_DML.sql	1,309 KB	INSERT statements for all tables (15,712 rows). Run second.
MUSE_Phase2_Verify.sql	7 KB	Read-only checks: table and constraint counts, rows per table, four join queries over the data.
MUSE_Phase2_Constraint_Tests.sql	3 KB	19 tests that try to break the constraints; changes that succeed are rolled back.

The scripts are generated from the live project database by backend/scripts/build-phase2-sql.js, so the DDL is exactly the schema the MUSE web application uses. They use their own database, muse_db, so the application's muse database is never touched.

1.1 How to run in MySQL Workbench 8.0 CE

1. Open Workbench and connect to your local server as root (or any account that can create databases).
2. **File → Open SQL Script…** → MUSE_Phase2_DDL.sql → click the lightning bolt (Execute all). Refresh the *Schemas* panel: muse_db appears with 64 tables.
3. Open MUSE_Phase2_DML.sql and execute it the same way (about a second on the test machine).
4. Open MUSE_Phase2_Verify.sql and execute it; each result tab shows one check.
5. Open MUSE_Phase2_Constraint_Tests.sql. Most of its statements are *meant* to fail, and Workbench stops at the first error, so either run them one at a time (cursor on the statement, **Ctrl + Enter**) or switch on *Continue on SQL Script Error* in the SQL editor toolbar before Execute all.

The DDL starts with DROP DATABASE IF EXISTS muse_db, so running DDL then DML again always gives a clean copy. Requirements: MySQL Server 8.0.16 or newer (CHECK constraints are enforced from 8.0.16).

2. Schema created by the DDL

Object	Count	Notes
Tables	64	All InnoDB, character set utf8mb4, collation utf8mb4_0900_ai_ci
Columns	523	Including 17 generated (derived) columns, computed by MySQL
Primary keys	64	One per table; surrogate AUTO_INCREMENT keys for entities, composite keys for associative tables
Foreign keys	124	69 CASCADE, 32 SET NULL, 23 RESTRICT on delete
Unique keys	39	Candidate (alternate) keys, e.g. users.username, users.email, moods.name
CHECK constraints	71	Value ranges, formats (regular expressions) and rules between columns
Indexes	222	64 primary, 39 unique, 115 ordinary, 4 FULLTEXT (ngram parser, for Indian-language titles)

2.1 Design rules used in the DDL

- **NOT NULL by default.** A column is nullable only when "unknown" or "not applicable" is a real state (e.g. songs.album_id for a single, listening_history.ended_at while a song is still playing).
- **Smallest exact type.** Lookup keys are TINYINT / SMALLINT UNSIGNED; high-volume event keys are BIGINT UNSIGNED; scores are DECIMAL (no rounding errors); fixed lists are ENUM.
- **Foreign-key actions chosen per relationship.** CASCADE where the child cannot exist alone (a playlist's songs); SET NULL where the child stays meaningful (a song whose album is removed); RESTRICT where deleting the parent would destroy history (a role still in use, a listener with a weekly MUSE World).
- **Rules in the database, not only in the app.** CHECK constraints keep ratings in 1–5, confidences in (0, 1], hex colours valid, end times after start times, and stop a play being both completed and skipped.
- **Derived values as generated columns.** For example albums.normalized_title (lower-cased title without punctuation, used for duplicate detection) is computed by MySQL and cannot be written directly.
- **Naming.** pk = <entity>_id; constraints are named fk_…, uq_…, chk_…, indexes idx_…, so every error message names the rule that was broken.

2.2 Tables by area

Cols = columns · FK = foreign keys · UK = unique keys · Gen. = generated columns · Rows = sample rows loaded by the DML.

Users — 11 tables: accounts, roles and permissions, profiles, languages and regions, sign-in sessions, audit log

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
users	11	user_id	1	2	2	4		13
roles	3	role_id	0	1	0	2		3
permissions	2	permission_code	0	0	0	1		9
role_permissions	2	role_id, permission_code	2	0	0	2		14
auth_sessions	10	session_id	1	1	2	4		0
user_languages	3	user_id, language_id	2	1	1	3		26
audit_logs	9	audit_id	1	0	0	4		1
schema_migrations	5	version	0	0	1	1		5
user_profiles	7	user_id	2	0	0	2	1	13
regions	4	region_code	1	1	1	3		14
languages	5	language_id	0	2	1	3		12
Total	**61**		**10**	**8**	**8**	**29**	**1**	**110**

Music Catalogue — 18 tables: artists, albums, films, songs, genres, music sources and the import pipeline

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
artists	11	artist_id	2	1	1	5	1	31
artist_aliases	5	alias_id	1	1	0	4	1	22
user_artist_preferences	14	user_id, artist_id	3	0	1	4		273
albums	9	album_id	2	0	3	5	1	32
album_artists	3	album_id, artist_id	2	0	0	2		32
films	4	film_id	1	1	1	3		4
songs	18	song_id	4	2	6	8	2	104
song_artists	4	song_id, artist_id, role	2	0	0	2		18
song_media	10	media_id	2	1	1	4		98
music_sources	16	source_id	1	1	1	3		9
source_records	21	record_id	6	1	2	9		135
pipeline_jobs	19	job_id	3	1	2	8		4
pipeline_job_logs	7	log_id	2	0	0	4		15
review_queue	13	review_id	4	1	1	7	1	0
trend_snapshots	13	snapshot_id	2	1	0	4	1	2,109
genres	5	genre_id	1	1	1	3		23
user_genre_preferences	14	user_id, genre_id	3	0	1	4		182
song_genres	4	song_id, genre_id	2	1	0	3	1	141
Total	**190**		**43**	**13**	**21**	**82**	**8**	**3,232**

Mood Intelligence — 15 tables: moods, mood keywords and contexts, audio features, AI mood tags, model versions, user mood-match ratings

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
moods	13	mood_id	1	1	3	3		17
mood_categories	4	category_id	0	2	0	3		4
mood_keywords	4	keyword_id	1	1	2	3		169
contexts	14	context_id	0	1	3	2		10
context_keywords	2	phrase	1	0	1	2		73
context_moods	3	context_id, mood_id	2	0	1	2		32
user_mood_preferences	10	user_id, mood_id	3	0	1	4		180
user_context_preferences	7	user_id, context_id	2	0	1	2		27

song_moods	6	song_id, mood_id, model_id	4	0	1	4		726
audio_features	16	feature_id	2	2	4	5	1	208
mood_segments	11	segment_id	3	1	2	4		407
song_mood_curations	7	song_id, mood_id	3	0	1	3	1	0
intelligence_models	8	model_id	0	2	0	3	1	9
model_parameters	6	model_id, param_code	1	0	1	1		48
song_mood_ratings	7	user_id, song_id, mood_id	3	0	2	3	1	0
Total	**118**		**26**	**10**	**23**	**44**	**4**	**1,910**

Listening History — 9 tables: listening sessions, plays, playback events, ratings, likes, dismissals, searches, incognito sessions

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
listening_sessions	10	session_id	2	1	1	4		497
incognito_sessions	8	incognito_id	2	1	2	5	1	2
incognito_events	7	event_id	2	0	1	3		8
search_history	6	search_id	1	0	0	3		33
listening_history	13	play_id	3	0	3	7		2,634
playback_events	6	event_id	1	0	1	2		5,942
ratings	5	user_id, song_id	2	0	1	2	1	146
user_likes	3	user_id, song_id	2	0	0	3		27
song_dismissals	4	user_id, song_id	3	0	0	3		0
Total	**62**		**18**	**2**	**9**	**32**	**2**	**9,289**

Recommendations — 9 tables: recommendation batches, ranked songs with reasons, weekly MUSE Worlds and their feedback

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
recommendation_batches	20	batch_id	6	0	2	7		11
recommendation_batch_moods	3	batch_id, mood_id	2	0	1	2		39
weekly_worlds	18	world_id	7	3	2	9	1	11
world_archetypes	7	archetype_code	0	1	1	2		13
archetype_moods	3	archetype_code, mood_id	2	0	1	2		50
world_feedback	5	user_id, world_id	2	0	0	2	1	0
recommendation_items	13	batch_id, song_id	2	1	1	3		198
recommendation_reasons	5	batch_id, song_id, reason_code	2	0	0	2		770
reason_types	3	reason_code	0	0	0	1		16
Total	**77**		**23**	**5**	**8**	**30**	**2**	**1,108**

Playlists — 2 tables: user playlists and their songs

Table	Cols	Primary key	FK	UK	CHECK	Indexes	Gen.	Rows
playlists	11	playlist_id	2	1	2	3		8
playlist_songs	4	playlist_id, song_id	2	0	0	2		55
Total	**15**		**4**	**1**	**2**	**5**	**0**	**63**

3. Constraint catalogue

3.1 Foreign keys (124)

In InnoDB, `NO ACTION` behaves exactly like `RESTRICT`, so it is shown as RESTRICT.

Child table	Column(s)	References	ON DELETE	ON UPDATE	Constraint
album_artists	album_id	albums(album_id)	CASCADE	RESTRICT	fk_aa_album
album_artists	artist_id	artists(artist_id)	CASCADE	RESTRICT	fk_aa_artist
albums	film_id	films(film_id)	RESTRICT	RESTRICT	fk_albums_film
albums	language_id	languages(language_id)	SET NULL	CASCADE	fk_albums_lang
archetype_moods	archetype_code	world_archetypes(archetype_code)	CASCADE	CASCADE	fk_am_arch
archetype_moods	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_am_mood
artist_aliases	artist_id	artists(artist_id)	CASCADE	RESTRICT	fk_aliases_artist
artists	primary_language_id	languages(language_id)	SET NULL	CASCADE	fk_artists_lang
artists	region_code	regions(region_code)	SET NULL	CASCADE	fk_artists_region
audio_features	model_id	intelligence_models(model_id)	RESTRICT	RESTRICT	fk_af_model
audio_features	song_id	songs(song_id)	CASCADE	RESTRICT	fk_af_song
audit_logs	actor_user_id	users(user_id)	SET NULL	RESTRICT	fk_audit_actor
auth_sessions	user_id	users(user_id)	CASCADE	RESTRICT	fk_session_user
context_keywords	context_id	contexts(context_id)	CASCADE	RESTRICT	fk_ck_ctx
context_moods	context_id	contexts(context_id)	CASCADE	RESTRICT	fk_cm_ctx
context_moods	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_cm_mood
films	language_id	languages(language_id)	RESTRICT	CASCADE	fk_films_lang
genres	parent_genre_id	genres(genre_id)	SET NULL	CASCADE	fk_genres_parent
incognito_events	incognito_id	incognito_sessions(incognito_id)	CASCADE	RESTRICT	fk_ie_session
incognito_events	song_id	songs(song_id)	CASCADE	RESTRICT	fk_ie_song
incognito_sessions	current_mood_id	moods(mood_id)	SET NULL	RESTRICT	fk_incog_mood
incognito_sessions	user_id	users(user_id)	CASCADE	RESTRICT	fk_incog_user
listening_history	batch_id	recommendation_batches(batch_id)	SET NULL	RESTRICT	fk_lh_batch
listening_history	session_id, user_id	listening_sessions(session_id, user_id)	CASCADE	RESTRICT	fk_lh_session_user
listening_history	song_id	songs(song_id)	CASCADE	RESTRICT	fk_lh_song
listening_sessions	context_id	contexts(context_id)	SET NULL	RESTRICT	fk_ls_ctx
listening_sessions	user_id	users(user_id)	CASCADE	RESTRICT	fk_ls_user
model_parameters	model_id	intelligence_models(model_id)	CASCADE	RESTRICT	fk_mp_model
mood_keywords	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_mk_mood
mood_segments	model_id	intelligence_models(model_id)	RESTRICT	RESTRICT	fk_ms_model
mood_segments	mood_id	moods(mood_id)	RESTRICT	RESTRICT	fk_ms_mood
mood_segments	song_id	songs(song_id)	CASCADE	RESTRICT	fk_ms_song
moods	category_id	mood_categories(category_id)	RESTRICT	CASCADE	fk_moods_category
music_sources	fallback_source_id	music_sources(source_id)	SET NULL	RESTRICT	fk_sources_fallback
pipeline_job_logs	job_id	pipeline_jobs(job_id)	CASCADE	RESTRICT	fk_logs_job
pipeline_job_logs	record_id	source_records(record_id)	SET NULL	RESTRICT	fk_logs_record
pipeline_jobs	parent_job_id	pipeline_jobs(job_id)	SET NULL	RESTRICT	fk_jobs_parent
pipeline_jobs	source_id	music_sources(source_id)	SET NULL	RESTRICT	fk_jobs_source
pipeline_jobs	requested_by	users(user_id)	SET NULL	RESTRICT	fk_jobs_user
playback_events	play_id	listening_history(play_id)	CASCADE	RESTRICT	fk_pe_play
playlist_songs	playlist_id	playlists(playlist_id)	CASCADE	RESTRICT	fk_ps_playlist
playlist_songs	song_id	songs(song_id)	CASCADE	RESTRICT	fk_ps_song
playlists	source_batch_id	recommendation_batches(batch_id)	SET NULL	RESTRICT	fk_pl_batch
playlists	user_id	users(user_id)	CASCADE	RESTRICT	fk_pl_user

ratings	song_id	songs(song_id)	CASCADE	RESTRICT	fk_ratings_song
ratings	user_id	users(user_id)	CASCADE	RESTRICT	fk_ratings_user
recommendation_batch_moods	batch_id	recommendation_batches(batch_id)	CASCADE	RESTRICT	fk_rbm_batch
recommendation_batch_moods	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_rbm_mood
recommendation_batches	context_id	contexts(context_id)	SET NULL	RESTRICT	fk_rb_ctx
recommendation_batches	incognito_id	incognito_sessions(incognito_id)	CASCADE	RESTRICT	fk_rb_incognito
recommendation_batches	language_id	languages(language_id)	SET NULL	RESTRICT	fk_rb_lang
recommendation_batches	model_id	intelligence_models(model_id)	RESTRICT	RESTRICT	fk_rb_model
recommendation_batches	seed_song_id	songs(song_id)	SET NULL	RESTRICT	fk_rb_seed
recommendation_batches	user_id	users(user_id)	CASCADE	RESTRICT	fk_rb_user
recommendation_items	batch_id	recommendation_batches(batch_id)	CASCADE	RESTRICT	fk_ri_batch
recommendation_items	song_id	songs(song_id)	CASCADE	RESTRICT	fk_ri_song
recommendation_reasons	batch_id, song_id	recommendation_items(batch_id, song_id)	CASCADE	RESTRICT	fk_rr_item
recommendation_reasons	reason_code	reason_types(reason_code)	RESTRICT	CASCADE	fk_rr_type
regions	primary_language_id	languages(language_id)	SET NULL	CASCADE	fk_regions_language
review_queue	job_id	pipeline_jobs(job_id)	SET NULL	RESTRICT	fk_rq_job
review_queue	record_id	source_records(record_id)	CASCADE	RESTRICT	fk_rq_record
review_queue	candidate_song_id	songs(song_id)	SET NULL	RESTRICT	fk_rq_song
review_queue	decided_by	users(user_id)	SET NULL	RESTRICT	fk_rq_user
role_permissions	permission_code	permissions(permission_code)	CASCADE	CASCADE	fk_rp_perm
role_permissions	role_id	roles(role_id)	CASCADE	CASCADE	fk_rp_role
search_history	user_id	users(user_id)	CASCADE	RESTRICT	fk_sh_user
song_artists	artist_id	artists(artist_id)	CASCADE	RESTRICT	fk_sa_artist
song_artists	song_id	songs(song_id)	CASCADE	RESTRICT	fk_sa_song
song_dismissals	batch_id	recommendation_batches(batch_id)	SET NULL	RESTRICT	fk_dismiss_batch
song_dismissals	song_id	songs(song_id)	CASCADE	RESTRICT	fk_dismiss_song
song_dismissals	user_id	users(user_id)	CASCADE	RESTRICT	fk_dismiss_user
song_genres	genre_id	genres(genre_id)	RESTRICT	CASCADE	fk_sg_genre
song_genres	song_id	songs(song_id)	CASCADE	RESTRICT	fk_sg_song
song_media	song_id	songs(song_id)	CASCADE	RESTRICT	fk_media_song
song_media	source_id	music_sources(source_id)	RESTRICT	RESTRICT	fk_media_source
song_mood_curations	curated_by	users(user_id)	SET NULL	RESTRICT	fk_smc_curator
song_mood_curations	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_smc_mood
song_mood_curations	song_id	songs(song_id)	CASCADE	RESTRICT	fk_smc_song
song_mood_ratings	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_smr_mood
song_mood_ratings	song_id	songs(song_id)	CASCADE	RESTRICT	fk_smr_song
song_mood_ratings	user_id	users(user_id)	CASCADE	RESTRICT	fk_smr_user
song_moods	feature_id	audio_features(feature_id)	CASCADE	RESTRICT	fk_sm_feature
song_moods	model_id	intelligence_models(model_id)	RESTRICT	RESTRICT	fk_sm_model
song_moods	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_sm_mood
song_moods	song_id	songs(song_id)	CASCADE	RESTRICT	fk_sm_song
songs	album_id	albums(album_id)	SET NULL	CASCADE	fk_songs_album
songs	primary_artist_id	artists(artist_id)	RESTRICT	CASCADE	fk_songs_artist
songs	language_id	languages(language_id)	RESTRICT	CASCADE	fk_songs_lang
songs	merged_into_song_id	songs(song_id)	RESTRICT	RESTRICT	fk_songs_merged
source_records	album_id	albums(album_id)	RESTRICT	RESTRICT	fk_sr_album
source_records	artist_id	artists(artist_id)	RESTRICT	RESTRICT	fk_sr_artist
source_records	film_id	films(film_id)	RESTRICT	RESTRICT	fk_sr_film
source_records	last_job_id	pipeline_jobs(job_id)	SET NULL	RESTRICT	fk_sr_job
source_records	song_id	songs(song_id)	RESTRICT	RESTRICT	fk_sr_song
source_records	source_id	music_sources(source_id)	RESTRICT	RESTRICT	fk_sr_source

trend_snapshots	job_id	pipeline_jobs(job_id)	CASCADE	RESTRICT	fk_trend_job
trend_snapshots	region_code	regions(region_code)	CASCADE	RESTRICT	fk_trend_region
user_artist_preferences	artist_id	artists(artist_id)	CASCADE	RESTRICT	fk_uap_artist
user_artist_preferences	model_id	intelligence_models(model_id)	SET NULL	RESTRICT	fk_uap_model
user_artist_preferences	user_id	users(user_id)	CASCADE	RESTRICT	fk_uap_user
user_context_preferences	context_id	contexts(context_id)	CASCADE	RESTRICT	fk_ucp_ctx
user_context_preferences	user_id	users(user_id)	CASCADE	RESTRICT	fk_ucp_user
user_genre_preferences	genre_id	genres(genre_id)	CASCADE	RESTRICT	fk_ugp_genre
user_genre_preferences	model_id	intelligence_models(model_id)	SET NULL	RESTRICT	fk_ugp_model
user_genre_preferences	user_id	users(user_id)	CASCADE	RESTRICT	fk_ugp_user
user_languages	language_id	languages(language_id)	CASCADE	CASCADE	fk_ul_lang
user_languages	user_id	users(user_id)	CASCADE	RESTRICT	fk_ul_user
user_likes	song_id	songs(song_id)	CASCADE	RESTRICT	fk_likes_song
user_likes	user_id	users(user_id)	CASCADE	RESTRICT	fk_likes_user
user_mood_preferences	model_id	intelligence_models(model_id)	SET NULL	RESTRICT	fk_ump_model
user_mood_preferences	mood_id	moods(mood_id)	CASCADE	RESTRICT	fk_ump_mood
user_mood_preferences	user_id	users(user_id)	CASCADE	RESTRICT	fk_ump_user
user_profiles	region_code	regions(region_code)	SET NULL	CASCADE	fk_up_region
user_profiles	user_id	users(user_id)	CASCADE	RESTRICT	fk_up_user
users	role_id	roles(role_id)	RESTRICT	CASCADE	fk_users_role
weekly_worlds	archetype_code	world_archetypes(archetype_code)	RESTRICT	CASCADE	fk_ww_arch
weekly_worlds	batch_id	recommendation_batches(batch_id)	RESTRICT	RESTRICT	fk_ww_batch
weekly_worlds	job_id	pipeline_jobs(job_id)	SET NULL	RESTRICT	fk_ww_job
weekly_worlds	language_id	languages(language_id)	SET NULL	RESTRICT	fk_ww_lang
weekly_worlds	primary_mood_id	moods(mood_id)	RESTRICT	RESTRICT	fk_ww_mood
weekly_worlds	region_code	regions(region_code)	SET NULL	CASCADE	fk_ww_region
weekly_worlds	user_id	users(user_id)	CASCADE	RESTRICT	fk_ww_user
world_feedback	user_id	users(user_id)	CASCADE	RESTRICT	fk_wf_user
world_feedback	world_id	weekly_worlds(world_id)	CASCADE	RESTRICT	fk_wf_world

3.2 CHECK constraints (71)

Table	Constraint	Condition
albums	chk_albums_colors	(regexp_like(`cover_color_1`,_utf8mb4\'^#[0-9A-Fa-f]{6}$\') and regexp_like(`cover_color_2`,_utf8mb4\'^#[0-9A-Fa-f]{6}$\'))
albums	chk_albums_film	((`film_id` is null) or (`album_type` = _utf8mb4\'SOUNDTRACK\'))
albums	chk_albums_year	(`release_year` between 1900 and 2100)
archetype_moods	chk_am_weight	((`weight` > 0) and (`weight` <= 1))
artists	chk_artists_year	((`formed_year` is null) or (`formed_year` between 1900 and 2100))
audio_features	chk_af_key	((`musical_key` <= 11) and (`mode` in (0,1)))
audio_features	chk_af_loudness	(`loudness_db` between -(60) and 0)
audio_features	chk_af_tempo	(`tempo_bpm` between 40 and 220)
audio_features	chk_af_unit	((`energy` between 0 and 1) and (`valence` between 0 and 1) and (`danceability` between 0 and 1) and (`acousticness` between 0 and 1) and (`instrumentalness` between 0 and 1) and (`speechiness` between 0 and 1))
auth_sessions	chk_session_revoke	((`revoked_at` is null) = (`revoke_reason` is null))
auth_sessions	chk_session_times	((`expires_at` > `created_at`) and (`last_seen_at` >= `created_at`))
context_keywords	chk_ck_lower	(`phrase` = lower(`phrase`))
context_moods	chk_cm_weight	((`weight` > 0) and (`weight` <= 1))
contexts	chk_ctx_code	regexp_like(`code`,_utf8mb4\'^[A-Z_]{2,20}$\',_utf8mb4\'c\')
contexts	chk_ctx_targets	((`target_energy` between 0 and 1) and (`target_valence` between 0 and 1) and (`target_danceability` between 0 and 1) and (`target_acousticness` between 0 and 1) and (`target_instrumentalness` between 0 and 1) and (`max_speechiness` between 0 and 1))
contexts	chk_ctx_tempo	((`tempo_min` >= 40) and (`tempo_max` <= 220) and (`tempo_min` < `tempo_max`))

films	chk_films_year	(`release_year` between 1900 and 2100)
genres	chk_genres_color	regexp_like(`color_hex`,_utf8mb4\'^#[0-9A-Fa-f]{6}$\')
incognito_events	chk_ie_flags	((`completed` <> 1) or (`skipped` <> 1))
incognito_sessions	chk_incog_state	((`status` = _utf8mb4\'ACTIVE\') = (`ended_at` is null))
incognito_sessions	chk_incog_times	((`expires_at` > `started_at`) and ((`ended_at` is null) or (`ended_at` >= `started_at`)))
languages	chk_languages_iso	regexp_like(`iso_code`,_utf8mb4\'^[a-z]{2,3}$\',_utf8mb4\'c\')
listening_history	chk_lh_end	((`ended_at` is null) or (`ended_at` >= `started_at`))
listening_history	chk_lh_flags	((`completed` <> 1) or (`skipped` <> 1))
listening_history	chk_lh_pct	(`completion_pct` between 0 and 100)
listening_sessions	chk_ls_state	(((`status` = _utf8mb4\'OPEN\') and (`ended_at` is null)) or ((`status` = _utf8mb4\'CLOSED\') and (`ended_at` is not null) and (`ended_at` >= `started_at`)))
model_parameters	chk_mp_range	((`min_value` <= `max_value`) and (`param_value` between `min_value` and `max_value`))
mood_keywords	chk_mk_lower	(`phrase` = lower(`phrase`))
mood_keywords	chk_mk_weight	((`weight` > 0) and (`weight` <= 1))
mood_segments	chk_ms_span	(`start_sec` < `end_sec`)
mood_segments	chk_ms_vals	((`confidence` > 0) and (`confidence` <= 1) and (`energy` between 0 and 1) and (`valence` between 0 and 1))
moods	chk_moods_color	regexp_like(`color_hex`,_utf8mb4\'^#[0-9A-Fa-f]{6}$\')
moods	chk_moods_targets	((`target_energy` between 0 and 1) and (`target_valence` between 0 and 1) and (`target_danceability` between 0 and 1) and (`target_acousticness` between 0 and 1) and (`target_instrumentalness` between 0 and 1))
moods	chk_moods_tempo	((`tempo_min` >= 40) and (`tempo_max` <= 220) and (`tempo_min` < `tempo_max`))
music_sources	chk_sources_attrib	((`requires_attribution` = 0) or (`attribution_text` is not null))
pipeline_jobs	chk_jobs_retry	(`retry_count` <= (`max_retries` + 1))
pipeline_jobs	chk_jobs_times	((`completed_at` is null) or (`started_at` is null) or (`completed_at` >= `started_at`))
playback_events	chk_pe_seek	((`event_type` = _utf8mb4\'SEEK\') = (`seek_to_sec` is not null))
playlists	chk_pl_name	(char_length(trim(`name`)) >= 1)
playlists	chk_pl_smart	((`kind` = _utf8mb4\'SMART\') = (`smart_type` is not null))
ratings	chk_ratings_value	(`rating` between 1 and 5)
recommendation_batch_moods	chk_rbm_weight	((`weight` > 0) and (`weight` <= 1))
recommendation_batches	chk_rb_targets	((`target_energy` between 0 and 1) and (`target_valence` between 0 and 1) and (`target_danceability` between 0 and 1) and (`target_acousticness` between 0 and 1) and (`target_instrumentalness` between 0 and 1) and (`max_speechiness` between 0 and 1))
recommendation_batches	chk_rb_tempo	(`tempo_min` < `tempo_max`)
recommendation_items	chk_ri_score	(`final_score` between 0 and 1)
regions	chk_regions_code	regexp_like(`region_code`,_utf8mb4\'^[A-Z]{2}(-[A-Z]{2,3})?$\',_utf8mb4\'c\')
review_queue	chk_rq_decided	((`status` = _utf8mb4\'PENDING\') = (`decided_at` is null))
schema_migrations	chk_mig_version	regexp_like(`version`,_utf8mb4\'^[0-9]{4}$\')
song_media	chk_media_uri	((`media_type` = _utf8mb4\'GENERATED_SKETCH\') or (`uri` is not null))
song_mood_curations	chk_smc_conf	(((`action` = _utf8mb4\'ADD\') and (`confidence` > 0) and (`confidence` <= 1)) or ((`action` = _utf8mb4\'REMOVE\') and (`confidence` is null)))
song_mood_ratings	chk_smr_pct	((`match_percentage` >= 0) and (`match_percentage` <= 100))
song_mood_ratings	chk_smr_review	((`review_text` is null) or (char_length(trim(`review_text`)) > 0))
song_moods	chk_sm_conf	((`confidence` > 0) and (`confidence` <= 1))
songs	chk_songs_duration	(`duration_sec` between 30 and 1800)
songs	chk_songs_explicit	((`explicit` = 0) or (`content_rating` <> _utf8mb4\'ALL\'))
songs	chk_songs_isrc	((`isrc` is null) or regexp_like(`isrc`,_utf8mb4\'^[A-Z]{2}[A-Z0-9]{3}[0-9]{7}$\',_utf8mb4\'c\'))
songs	chk_songs_merge	((`status` = _utf8mb4\'MERGED\') = (`merged_into_song_id` is not null))
songs	chk_songs_pop	(`popularity` <= 100)
songs	chk_songs_year	(`release_year` between 1900 and 2100)
source_records	chk_sr_entity	(((`entity_type` = _utf8mb4\'SONG\') and (`artist_id` is null) and (`album_id` is null) and (`film_id` is null)) or ((`entity_type` = _utf8mb4\'ARTIST\') and (`song_id` is null) and (`album_id` is null) and (`film_id` is null)) or ((`entity_type` = _utf8mb4\'ALBUM\') and (`song_id` is null) and (`artist_id` is null) and (`film_id` is null)) or ((`entity_type` = _utf8mb4\'FILM\') and (`song_id` is null) and (`artist_id` is null) and (`album_id` is null)))
source_records	chk_sr_scores	(((`match_confidence` is null) or (`match_confidence` between 0 and 1)) and ((`quality_score` is null) or (`quality_score` between 0 and 1)))

user_artist_preferences	chk_uap	((`onboarding_weight` between 0 and 1) and (`blended_score` between 0 and 1))
user_context_preferences	chk_ucp	((`onboarding_weight` between 0 and 1) and (`blended_score` between 0 and 1))
user_genre_preferences	chk_ugp	((`onboarding_weight` between 0 and 1) and (`blended_score` between 0 and 1))
user_languages	chk_ul_priority	(`priority` between 1 and 10)
user_mood_preferences	chk_ump	((`onboarding_weight` between 0 and 1) and (`blended_score` between 0 and 1))
users	chk_users_email	(`email` like _utf8mb4\'%_@_%._%\')
users	chk_users_username	regexp_like(`username`,_utf8mb4\'^[a-z0-9_]{3,30}$\',_utf8mb4\'c\')
weekly_worlds	chk_ww_expiry	(`expires_at` > `generated_at`)
weekly_worlds	chk_ww_week	(dayofweek(`week_start`) = 2)
world_archetypes	chk_arch_palette	(json_length(`palette`) between 3 and 6)

4. Sample data (DML)

The DML inserts the MUSE demo data set: 15,712 rows. It is the same data the web application's own setup (`npm run db:setup`) creates — demo users, a catalogue of 104 songs by 31 artists in 12 languages, mood analysis for every song, 2,634 plays in 497 listening sessions, recommendations with reasons, and playlists. Generated columns are not in the INSERTs; MySQL computes them. The script switches `FOREIGN_KEY_CHECKS` off while loading, because some tables refer to themselves (genre → parent genre, pipeline job → parent job), and on again at the end; section 5.3 shows that no row breaks a foreign key.

Largest tables: `playback_events` (5,942), `listening_history` (2,634), `trend_snapshots` (2,109), `recommendation_reasons` (770), `song_moods` (726), `listening_sessions` (497). Empty on purpose: `auth_sessions`, `review_queue`, `song_dismissals`, `song_mood_curations`, `song_mood_ratings`, `world_feedback` — they fill only while the web application is used (sign-in sessions, mood-match ratings, review queue …); the design does not invent such rows.

Password hashes in `users` are bcrypt hashes of demo passwords; no real personal data is included.

users (with role and profile)

user_id	username	email	role	display_name	is_active	created_at
1	admin	admin@muse.dev	Administrator	Asha Rao	1	2026-08-17 15:32:25
2	kavya	kavya@muse.dev	Curator	Kavya Reddy	1	2026-08-18 15:32:25
3	aarav	aarav@muse.dev	Listener	Aarav Varma	1	2026-08-19 15:32:25
4	meera	meera@muse.dev	Listener	Meera Iyer	1	2026-08-20 15:32:25
5	rohan	rohan@muse.dev	Listener	Rohan Kulkarni	1	2026-08-21 15:32:25
6	sana	sana@muse.dev	Listener	Sana Qureshi	1	2026-08-22 15:32:25

songs (with artist, album, language)

song_id	title	artist	album	duration_sec	language	release_year
1	Vennela Vaana	Varsha Kondapalli	Vennela Vaana	228	Telugu	2022
2	Gunde Chappudu	Varsha Kondapalli	Vennela Vaana	239	Telugu	2022
3	Kotha Kotha Ga	Varsha Kondapalli	Vennela Vaana	206	Telugu	2022
4	Neeli Megham	Varsha Kondapalli	Vennela Vaana	295	Telugu	2022
5	Chinukulu	Varsha Kondapalli	NULL	285	Telugu	2024
6	Vennela Vaana	Varsha Kondapalli	NULL	298	Telugu	2023
7	Godari Gaali	Godavari Echoes	River Town Radio	238	Telugu	2021
8	Palleturi Daari	Godavari Echoes	River Town Radio	200	Telugu	2021

moods (target audio profile of each mood)

mood_id	name	category	color_hex	target_energy	target_valence	tempo_min	tempo_max
1	Happy	Bright & lively	#F6C453	0.720	0.850	100.0	135.0
2	Euphoric	Bright & lively	#FFB547	0.900	0.900	118.0	150.0
3	Energetic	Bright & lively	#FF8A4C	0.900	0.650	118.0	165.0
4	Motivational	Bright & lively	#F28C28	0.840	0.720	110.0	150.0
5	Confident	Bright & lively	#B28DFF	0.740	0.600	85.0	125.0
6	Adventurous	Bright & lively	#3FB8AF	0.720	0.600	100.0	145.0
7	Romantic	Warm & easy	#E86A92	0.400	0.620	70.0	110.0
8	Calm	Warm & easy	#7CC6B5	0.250	0.550	60.0	100.0

song_moods (mood tags from the active model)

song	mood	model	confidence
Vennela Vaana	Romantic	mood-rules 2.0	0.936
Vennela Vaana	Relaxed	mood-rules 2.0	0.836
Vennela Vaana	Nostalgic	mood-rules 2.0	0.701
Vennela Vaana	Dreamy	mood-rules 2.0	0.624
Gunde Chappudu	Happy	mood-rules 2.0	0.955

Gunde Chappudu	Confident	mood-rules 2.0	0.721
Gunde Chappudu	Motivational	mood-rules 2.0	0.718
Gunde Chappudu	Euphoric	mood-rules 2.0	0.704

listening_history (plays)

play_id	username	song	origin	started_at	listened_sec	completion_pct	completed	skipped
1	aarav	Gelupu Gelupu	HOME	2026-08-29 19:27:00	242	100.00	1	0
2	aarav	Dhol Di Awaaz	SEARCH	2026-08-29 19:31:07	209	100.00	1	0
3	aarav	Rowdy Rhythm	SEARCH	2026-08-29 19:34:43	195	100.00	1	0
4	aarav	Rowdy Rhythm	CONTEXT	2026-08-29 19:38:02	195	100.00	1	0
5	aarav	Rowdy Rhythm	PLAYLIST	2026-08-29 19:41:24	120	61.54	0	0
6	aarav	Kinetic Bloom	PLAYLIST	2026-08-29 19:43:28	252	100.00	1	0
7	aarav	Kinetic Bloom	PLAYLIST	2026-08-29 19:47:42	20	7.94	0	1
8	aarav	Raatbhar	LIBRARY	2026-08-30 07:00:00	232	100.00	1	0

playlists

playlist_id	name	owner	songs
1	Leg Day Arsenal	aarav	7
2	Hyderabad Nights	aarav	9
3	Deep Work Fuel	meera	5
4	Old Letters	rohan	5
5	Sangeet Night	sana	7
6	Houseboat at 2AM	leo	8
7	Rainy Tram	ishita	8
8	Sunday Morning Telugu	priya	6

5. Testing

Test machine: MySQL 8.0.46-0ubuntu0.24.04.4 with the default strict `sql_mode`:
`IGNORE_SPACE,ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION`

The DDL ran in 1.6 s and the DML in 0.6 s, both without errors. All results below come from `backend/scripts/test-phase2-sql.js`, which loads the two scripts into `muse_db` and compares it with the application database `muse` built by the project's migrations.

5.1 Structure: DDL result vs. the application schema

Compared from information_schema	Items	Differences
Columns (name, position, type, NULL, default, extra, generation expression, collation)	523	0
Index columns (PRIMARY, UNIQUE, ordinary, FULLTEXT)	318	0
CHECK constraints (name, table, condition)	71	0
Foreign keys (columns, parent, ON DELETE, ON UPDATE)	126	0
Tables (name, engine, collation)	64	0

5.2 Data: DML result vs. the application data

Check	Result
Row count of every table	64 tables compared, 0 different
`CHECKSUM TABLE` of every table (all column values)	64 tables compared, 0 different
Generated columns	Recomputed by MySQL from the inserted data; included in the checksums above, so they match the application values

5.3 Referential integrity

For each of the 124 foreign keys, a query counted child rows whose non-NULL key value has no matching parent row. Result: **0 orphan rows**.

5.4 Constraint tests

Each test tries to store something the design forbids, or checks a referential action. 15 statements were rejected by MySQL with the error shown. Tests T15–T18 must succeed: each is passed only if its before/after values are as expected, and their changes are rolled back. After all tests, the checksums of every table were compared again with `muse`: no data was changed.

ID	What is tested · statement	Actual MySQL result	Result
T01	PRIMARY KEY: same user rates the same song twice `INSERT INTO ratings (user_id, song_id, rating) VALUES (3, 3, 4)`	**Rejected** — ERROR 1062 (23000): Duplicate entry '3-3' for key 'ratings.PRIMARY'	Pass
T02	UNIQUE: username already taken `INSERT INTO users (username, email, password_hash, role_id) VALUES ('kavya', 'new.user@example.com', REPEAT('x', 60), 2)`	**Rejected** — ERROR 1062 (23000): Duplicate entry 'kavya' for key 'users.uq_users_username'	Pass
T03	FOREIGN KEY: rating for a song that does not exist `INSERT INTO ratings (user_id, song_id, rating) VALUES (3, 999999, 4)`	**Rejected** — ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`muse_db`.`ratings`, CONSTRAINT `fk_ratings_song` FOREIGN KEY (`song_id`) REFERENCES `songs` (`song_id`) ON DELETE CASCADE)	Pass
T04	CHECK: rating outside 1-5 `UPDATE ratings SET rating = 6 WHERE user_id = 3 AND song_id = 3`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_ratings_value' is violated.	Pass
T05	CHECK: username with capitals, spaces and symbols `INSERT INTO users (username, email, password_hash, role_id) VALUES ('Bad Name!', 'bad.name@example.com', REPEAT('x', 60), 2)`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_users_username' is violated.	Pass
T06	CHECK: email without @ `INSERT INTO users (username, email, password_hash, role_id) VALUES ('new_user', 'not-an-email', REPEAT('x', 60), 2)`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_users_email' is violated.	Pass
T07	NOT NULL: mood without a name `INSERT INTO moods (category_id, name, emoji, color_hex, description, target_energy, target_valence, target_danceability, target_acousticness, target_instrumentalness, tempo_min, tempo_max) VALUES (1, NULL, 'x', '#123456', 'test', 0.5, 0.5, 0.5, 0.5, 0.5, 80, 120)`	**Rejected** — ERROR 1048 (23000): Column 'name' cannot be null	Pass
T08	CHECK: a play cannot be both completed and skipped `UPDATE listening_history SET completed = 1, skipped = 1 WHERE play_id = 1`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_lh_flags' is violated.	Pass

T09	CHECK: mood confidence above 1 `UPDATE song_moods SET confidence = 1.5 WHERE song_id = 2 AND mood_id = 1 AND model_id = 3`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_sm_conf' is violated.	Pass
T10	CHECK: mood colour not a hex code `UPDATE moods SET color_hex = 'blue' WHERE mood_id = 1`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_moods_color' is violated.	Pass
T11	CHECK: tempo_min must be below tempo_max `UPDATE moods SET tempo_min = 150, tempo_max = 100 WHERE mood_id = 1`	**Rejected** — ERROR 3819 (HY000): Check constraint 'chk_moods_tempo' is violated.	Pass
T12	ENUM: origin value not in the list `UPDATE listening_history SET origin = 'RADIO' WHERE play_id = 1`	**Rejected** — ERROR 1265 (01000): Data truncated for column 'origin' at row 1	Pass
T13	GENERATED COLUMN: cannot be written directly `UPDATE albums SET normalized_title = 'x' WHERE album_id = 1`	**Rejected** — ERROR 3105 (HY000): The value specified for generated column 'normalized_title' in table 'albums' is not allowed.	Pass
T14	ON DELETE RESTRICT: role still used by users `DELETE FROM roles WHERE role_id = 2`	**Rejected** — ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`muse_db`.`users`, CONSTRAINT `fk_users_role` FOREIGN KEY (`role_id`) REFERENCES `roles` (`role_id`) ON DELETE RESTRICT ON UPDATE CASCADE)	Pass
T15	ON DELETE CASCADE: deleting a playlist removes its playlist_songs rows (rolled back) `DELETE FROM playlists WHERE playlist_id = 2`	✔ playlist_songs_before = 9 ✔ playlist_songs_after = 0	Pass
T15b	ON DELETE RESTRICT: a listener whose weekly MUSE World exists cannot be deleted `DELETE FROM users WHERE user_id = 3`	**Rejected** — ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`muse_db`.`weekly_worlds`, CONSTRAINT `fk_ww_batch` FOREIGN KEY (`batch_id`) REFERENCES `recommendation_batches` (`batch_id`) ON DELETE RESTRICT)	Pass
T16	ON DELETE SET NULL: deleting an album keeps its songs, album_id becomes NULL (rolled back) `DELETE FROM albums WHERE album_id = 1`	✔ songs_on_album_1_before = 4 ✔ songs_on_album_1_after = 0, songs_total = 104	Pass
T17	GENERATED COLUMN computed by MySQL `SELECT album_id, title, normalized_title FROM albums ORDER BY album_id LIMIT 3`	✔ album_id = 1, title = Vennela Vaana, normalized_title = vennela vaana	Pass
T18	valid insert succeeds (rolled back) `INSERT INTO users (username, email, password_hash, role_id) VALUES ('phase2_test', 'phase2.test@example.com', REPEAT('x', 60), 2)`	✔ user_id = 15, username = phase2_test, is_active = 1, failed_login_count = 0, created_at_defaulted = 1	Pass

6. Notes and limits

- Phase 2 covers tables, keys, constraints and indexes. The 20 views, 22 stored functions and procedures and 14 triggers of MUSE are created by `database/workbench/MUSE_Workbench_Setup.sql` (the full build) and belong to the query and programming phase.
- The tests show that the constraints listed here are enforced by MySQL; they do not prove that the design is free of every possible error, and the sample data is demo data, not a production workload.
- The FULLTEXT indexes use the `ngram` parser, which ships with MySQL 8.0; on other database systems these four indexes would need another full-text method.

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